

Nirma University
Institute of Technology
Semester End Examination (IR), January- 2025
Master of Computer Applications, Semester-I
6CS104CC22 Database Management System

Roll /
Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. All questions are compulsory.
2. Use section-wise separate answer book.
3. Figures to the right indicate full marks.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.

SECTION-I

- Q-1. Answer the following. [16]**
- (A) **CO-1** Discuss the roles and responsibilities of a Database administrator. (8)
BL-2
- (B) **CO-1** Illustrate referential integrity in a database with a suitable example. (8)
BL-3 Additionally, explain the concepts of alternate keys and artificial keys, providing appropriate examples to illustrate each.
- Q-2. Answer the following. [16]**
- (A) **CO-3** Explain full outer join and cross join with appropriate examples. (8)
BL-2 Illustrate how each join works using two tables: employees (employee_id, first_name) and departments (department_id, department_name).
- (B) **CO-2** Explain the use of the ALTER TABLE command in SQL with (8)
BL-4 appropriate syntax and examples. How is it used to:
- I. Add a new attribute to an existing relation
 - II. Drop an attribute from an existing relation
 - III. Rename a column in an existing relation
 - IV. Modify the data type of a column
 - V. Add a NOT NULL constraint to an existing column
- Q-3. Answer the following. [18]**
- (A) **CO-2** Illustrate the Concept of Normalization in DBMS and Detail the (8)
BL-3 Steps to Achieve the First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) with Examples.

- (B) **CO-3** Write a PL/SQL block for an explicit cursor to retrieve the order_id, (10)
BL-6 order_date, and total_amount from the ORDERS table for all orders with a total amount greater than 1000. Use the cursor to fetch and display the details of each order. Additionally, implement a mechanism to count the number of records fetched using the cursor. Explain the purpose of each step involved in using the explicit cursor and how the cursor is closed after use.

SECTION-II

- Q-4. Answer the following. [16]**
- (A) **CO-4** Explain the concepts of simple locking, pre-claiming locking, two- (8)
BL-2 phase locking, and strict two-phase locking.
- (B) **CO-4** Discuss Timestamp-based Protocols, focusing on how the Basic (8)
BL-2 Timestamp Ordering Protocol works. Additionally, briefly explore the advantages and disadvantages of the Timestamp Ordering Protocol.
- Q-5. Answer the following. [16]**
- (A) **CO-4** Explain the clustering index and the secondary index with suitable (8)
BL-2 examples.
- (B) **CO-4** Discuss the advantages and disadvantages of indexing in the (8)
BL-2 database management system.
- Q-6. Answer the following. [18]**
- (A) **CO-3** Create a PL/SQL block that uses a user-defined exception to check (8)
BL-6 for invalid age while inserting employee records into the EMPLOYEES table. The program should do the following:
- Accept the employee ID, name, and age as inputs.
 - If the employee's age is less than 18 or greater than 60, raise a user-defined exception called ex_invalid_age.
 - If the age is valid, insert the employee's details into the EMPLOYEES table.
 - Handle the exception and display an appropriate error message if the age is invalid.
- Assume and mention the data for the EMPLOYEES table.
- (B) **CO-3** What is the objective of a 'trigger' in DBMS? Create a row-level (10)
BL-6 trigger on the EMPLOYEES table that logs changes to the DEPARTMENT_ID of employees during UPDATE operations. The trigger should execute before any UPDATE operation on the DEPARTMENT_ID column. Assume and mention the necessary data for the EMPLOYEES table.
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